

GREENOCK MRCC, United Kingdom

WMO: 031380

Lat: 55.967N

Lon: 4.800W

Elev: 16

StdP: 14.69

Time Zone: 0.00 (EUW)

Period: 90-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.6	32.9	22.3	16.7	36.3	24.9	19.0	37.7	45.7	42.0	41.2	43.9	7.1	120	0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	71.8	61.9	69.0	60.8	66.4	59.0	63.9	69.7	62.1	66.5	60.7	65.0	8.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	81.7	66.1	60.0	77.4	64.7	58.6	73.6	63.2	29.1	70.3	27.9	66.4	26.8	65.0	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.6	31.4	27.4	DB	28.8	77.5	1.7	2.7	27.6	79.5	26.6	81.0	25.6	82.6	24.4	84.5	(4)
(5)			WB	26.7	67.1	2.5	3.6	24.8	69.7	23.4	71.8	21.9	73.8	20.1	76.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	42.2	42.0	44.3	47.9	52.9	56.6	59.8	59.8	56.0	50.8	46.5	42.3	(6)	
	DBStd	7.75	4.46	4.08	4.00	4.01	4.10	3.72	3.73	3.65	3.53	4.45	4.48	5.19	(7)	
	HDD50	1156	242	224	181	85	16	1	0	0	2	46	120	239	(8)	
	HDD65	5436	707	644	643	512	375	253	166	168	271	439	556	702	(9)	
	CDD50	1212	1	1	3	23	106	199	305	305	181	72	14	2	(10)	
	CDD65	15	0	0	0	0	0	1	6	7	1	0	0	0	(11)	
	CDH74	41	0	0	0	0	1	7	15	17	1	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	12.0	14.9	13.9	14.0	12.0	10.8	10.5	10.4	10.1	10.6	11.7	12.5	12.7	(14)	
(15) Precipitation	PrecAvg	60.3	7.0	5.2	5.2	3.6	3.3	3.3	3.9	4.6	4.8	6.5	6.3	6.7	(15)	
	PrecMax	74.6	10.5	10.0	9.2	5.9	6.5	5.9	6.5	10.0	8.3	10.8	12.0	11.6	(16)	
	PrecMin	45.9	1.9	1.4	2.4	1.9	0.7	1.4	1.2	1.3	1.4	2.1	2.4	1.8	(17)	
	PrecStd	7.8	2.3	2.5	1.7	1.0	1.5	1.2	1.3	2.2	1.9	2.6	2.3	3.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	53.2	55.2	64.4	70.7	74.1	76.3	76.3	69.2	61.9	56.7	54.5	(19)	
		MCWB	50.9	50.0	49.5	55.1	58.3	59.9	65.4	64.6	60.7	56.7	53.4	52.0	(20)	
	2%	DB	51.7	51.0	53.1	59.2	65.4	68.2	71.7	71.6	65.5	59.8	55.2	52.7	(21)	
		MCWB	49.4	48.0	48.9	51.5	56.2	58.6	62.7	61.8	58.8	56.0	52.3	50.2	(22)	
	5%	DB	50.3	49.8	51.5	55.8	62.2	65.0	69.1	68.1	63.1	58.2	54.2	51.2	(23)	
		MCWB	48.0	47.0	47.8	49.9	54.3	56.7	61.4	61.1	57.6	54.8	51.6	48.7	(24)	
	10%	DB	48.8	48.2	50.1	53.8	59.5	62.6	66.2	65.3	61.2	56.6	52.9	49.6	(25)	
		MCWB	46.3	45.3	46.8	48.9	52.8	55.8	59.5	59.4	56.7	53.3	50.4	47.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	50.3	52.1	56.7	60.6	63.2	67.1	66.2	62.1	58.1	54.5	52.6	(27)	
		MCDB	52.6	52.0	53.3	62.8	67.0	69.3	73.2	72.9	66.5	60.7	55.7	54.2	(28)	
	2%	WB	50.0	48.7	50.1	53.3	57.6	60.4	64.0	64.0	60.4	56.8	53.2	50.7	(29)	
		MCDB	51.4	50.6	52.0	56.9	63.6	66.0	70.2	69.9	63.7	59.2	54.6	52.5	(30)	
	5%	WB	48.3	47.3	48.7	51.3	55.6	58.4	62.2	62.2	59.0	55.4	52.0	49.0	(31)	
		MCDB	50.1	49.4	50.9	54.7	60.7	62.9	67.0	66.1	61.9	57.5	53.7	50.9	(32)	
	10%	WB	46.5	45.7	47.2	49.7	53.9	56.8	60.7	60.6	57.4	53.8	50.6	47.3	(33)	
		MCDB	48.5	47.9	49.6	53.1	58.3	60.9	65.1	64.6	60.5	56.2	52.7	49.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.5	7.4	8.9	10.3	9.4	9.0	9.0	8.1	6.5	6.1	5.6	(35)	
		MCDBR	7.4	7.6	8.8	13.0	15.8	14.4	14.3	14.5	10.9	7.7	6.9	7.6	(36)	
	5% WB	MCWBR	7.4	6.7	6.1	7.8	8.6	7.5	7.5	7.3	6.4	5.8	6.8	7.5	(37)	
		MCWBR	7.3	7.1	7.1	10.9	13.5	12.1	12.3	11.3	9.8	6.7	6.9	7.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.337	0.347	0.371	0.387	0.391	0.385	0.387	0.371	0.363	0.346	0.327	0.321	(40)	
		taud	2.196	2.296	2.277	2.276	2.314	2.364	2.366	2.422	2.441	2.413	2.288	2.173	(41)	
	Ebn at Noon		171	225	249	263	269	272	269	267	252	224	177	144	(42)	
		Edh at Noon	19	25	32	36	37	36	35	31	27	22	17	15	(43)	
(44) All-Sky Solar Radiation	RadAvg		145	354	699	1144	1479	1446	1384	1111	769	413	192	109	(44)	
	RadStd		21	45	56	90	138	159	139	90	61	56	19	14	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page